Sizing of a 16.4 V / 1100 A six-pulse thyristor rectifier

400 V line (360–440 V), transformer, line reactor with 5th/7th/11th traps, DC LC filter

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Abstract

This note sizes a single fully-controlled six-pulse thyristor bridge that must deliver 16.4 V at 1100 A into a resistive load from a 400 V line that may be as low as 360 V. The chain is: 400 V bus, series line reactor, tuned traps for the 5th, 7th and 11th harmonics, converter transformer, bridge, DC LC filter, load. The central result is the transformer secondary voltage: 17 V rms line-to-line at 400 V primary (15.2 V at 360 V, ratio 23.5), which gives a firing angle of 12° at 360 V, 28° at 400 V and 36° at 440 V. The transformer leakage inductance (2.1 μH per phase on the secondary, 1.2 mH on the primary) and winding resistance (0.29 mΩ per phase referred to the secondary, 0.25 Ω per primary delta phase) are estimated and their voltage drops included. The reactive demand never exceeds 17 kvar, the 11 kvar of the traps cover it, the current THD at the point of common coupling is below 4 % referred to the converter's own 36 A fundamental, and the load voltage ripple is below 2 % peak-to-peak with a 0.3 mH / 5 mF DC filter. The general relations and the design choices follow the companion note on the multi-voltage 1 MW supply [1], of which this unit is the 1-unit (B1) bridge.

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1 Scope and data

- Line: 400 V, 50 Hz, three-phase, ±10 % (360–440 V). Connection rated $S_n = 1$ MVA, fed by an MV/LV transformer with $u_k = 6$ % (assumed): $S_{sc} \approx 16.7$ MVA, $X_S = 9.6$ mΩ at 400 V.

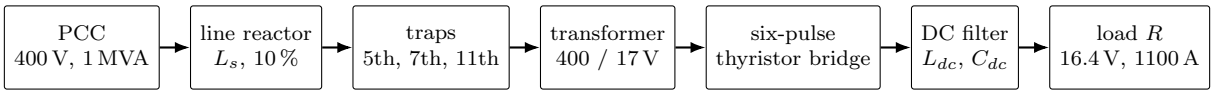


Figure 1: Power chain. The traps sit on the converter side of the line reactor, so that the reactor is between the filter bus and the grid.

- Load: resistive, $V_{out} = 16.4\,\mathrm{V}$, $I_d = 1050\,\mathrm{A}$ in service, 1100 A rated; $R = 14.9\,\mathrm{m\Omega}$. Power 17–18 kW.
- Targets: line-current THD $\leq 5\,\%$ at the PCC referred to the converter's own fundamental (the conservative reading; IEC 61000-3-12 and IEEE 519 refer harmonics to the rated or maximum demand current of the connection, 1443 A here, against which this unit is invisible); load voltage ripple $\leq 5\,\%$ peak-to-peak; firing angle low enough that no reactive compensation beyond the traps is needed.
- Bridge data: relative commutation reactance $e_x = 6\,\%$ and relative resistive drop $e_r = 2.7\,\%$ on the secondary rating (transformer plus wiring, Section 3.2), thyristor on-state drop $V_T = 1.3\,\mathrm{V}$ (two in series), minimum firing angle $\alpha_{min} = 10^\circ$ at the lowest line voltage and rated current.

2 Relations

For a six-pulse bridge with smooth DC current, secondary line-to-line voltage E_2 and firing angle α [2, 4]:

$$V_{d0} = \frac{3\sqrt{2}}{\pi} E_2 = 1.35\,E_2, \qquad V_d = V_{d0}\cos\alpha - d_x - d_r - 2V_T, \tag{1}$$

$$d_x = \frac{e_x}{2} V_{d0} \frac{I_d}{I_{d,r}}, \qquad d_r = e_r V_{d0} \frac{I_d}{I_{d,r}},$$

$$\cos\alpha - \cos(\alpha+u) = \frac{2d_x}{V_{d0}}, \qquad I_2 = 0.816\,I_d, \qquad I_1 = 0.78\,I_d, \qquad S_1 = V_{d0} I_d, \tag{2}$$

$$\cos\varphi_1 \simeq \frac{\cos\alpha + \cos(\alpha+u)}{2}, \qquad P \simeq V_d I_d, \qquad Q = S_1 \sin\varphi_1, \tag{3}$$

$$\frac{I_h}{I_1} = \frac{1}{h} \frac{\sqrt{A^2 + B^2 - 2AB\cos(2\alpha+u)}}{\cos\alpha - \cos(\alpha+u)}, \qquad h = 6k \pm 1, \tag{4}$$

$$A = \frac{\sin\frac{(h-1)u}{2}}{h-1}, \qquad B = \frac{\sin\frac{(h+1)u}{2}}{h+1},$$

$$\frac{V_n}{V_{d0}} = \frac{\sqrt{2}}{n^2-1}\sqrt{1 + (n^2-1)\sin^2\alpha}, \quad n = 6k \quad \text{(DC-side ripple, rms)}. \tag{5}$$

The resistive drop d_r is the IEC 60146-1-1 ohmic direct-voltage regulation: with R_k the total winding resistance per phase referred to the secondary (star equivalent), the DC current flows through two phases at a time and $d_r = 2R_k I_d$; expressed through the relative short-circuit resistance $e_r = \sqrt{3} R_k I_2 / E_2$ this is $d_r = 1.05\,e_r V_{d0} \approx e_r V_{d0}$.

3 Transformer secondary voltage

The bridge must deliver V_{out} at rated current with the line at its minimum, at $\alpha = \alpha_{min}$. From Eq. (1):

$$V_{d0,360} \geq \frac{V_{out} + 2V_T}{\cos\alpha_{min} - e_x/2 - e_r} = \frac{16.4 + 2.6}{0.985 - 0.030 - 0.027} = 20.5\ \mathrm{V}, \quad E_{2,360} = \frac{20.5}{1.35} = 15.2\ \mathrm{V}. \tag{6}$$

Referred to the nominal primary voltage this is $15.2 \times 400/360 = 16.8\,\text{V}$; the rated secondary is rounded up to

$$\boxed{E_2 = 17\ \text{V rms line-to-line at } 400\,\text{V primary, ratio } 400/17 = 23.5}$$

so that the 16.4 V output is reached at 360 V with $\alpha = 12°$ (the extra 1 % of voltage is absorbed by the angle). Without the resistive term the result would be 16.5 V: the winding resistance costs half a volt of output at 1100 A and cannot be neglected at this voltage level. Three remarks:

- The two thyristor drops (2.6 V) are 16 % of the output voltage and are added *before* the margin factor; treating them as a percentage would under-size the winding.
- $e_x = 6\,\%$ must include everything between the filter bus and the bridge, referred to the secondary. The line reactor of Section 5 is on the primary side of the traps and does not enter e_x: at the fundamental its drop is nearly cancelled by the capacitive current of the traps, and at the harmonics its job is to separate the filter bus from the grid, not to soften the commutation. If instead the reactor were placed on the secondary, e_x would rise to about 10 % and E_2 to 17.4 V.
- Primary taps $\pm 2 \times 2.5\,\%$ are not necessary here (the $\pm 10\,\%$ line range is covered by the angle with a maximum of 36°), but are cheap and allow centring the angle on the actual site voltage; if fitted, the transformer is specified 400 V $\pm 2 \times 2.5\,\%$ / 17 V.

3.1 Transformer data

- Three-phase, primary 400 V delta, secondary 17 V star (or delta; a single secondary has no phase-shift requirement). Delta primary to block triplen harmonics and give a return path for the zero-sequence magnetising current.
- Secondary current $I_2 = 0.816 \times 1100 = 898\,\text{A}$ rms, six-pulse waveform (120° blocks); rating $S_2 = \sqrt{3} \times 17 \times 898 = 26.4\,\text{kVA}$; primary current 38 A rms at 400 V (fundamental 36.5 A).
- Short-circuit reactance $e_x = 6\,\%$ (specified as 5–7 %) and resistance $e_r \approx 2.7\,\%$ on 26.4 kVA (Section 3.2); the very low secondary voltage (17 V at 898 A) means a foil or bar secondary with a few turns per phase and a leakage reactance dominated by the busbar layout, which is why e_x should be measured, not assumed.
- Converter-duty specification per IEC 61378-1 [3]: harmonic loss factor $F_{HL} = \sum_h (I_h/I_1)^2 h^2 \approx 5$ (up to the 49th, $u \approx$ 6–11°) applied to the eddy-loss share of the load losses; the DC component is zero for a bridge connection.
- Flux density chosen for 1.10 line voltage (440 V) plus the 4.4 % bus rise at no load caused by the traps through the line reactor (Section 5): the core must not saturate at $1.15\,V_n$.
- Inrush at 25 kVA is not an issue; a standard contactor with the traps switched together with the transformer is sufficient.

3.2 Leakage inductance and winding resistance

Leakage (short-circuit) inductance. The base impedance of the secondary is $Z_{b2} = E_2^2/S_2 = 17^2/26\,400 = 10.9\,\text{m}\Omega$ (star equivalent, per phase). With $e_x = 6\,\%$:

$$X_k = e_x Z_{b2} = 0.66\ \text{m}\Omega, \qquad L_k = \frac{X_k}{\omega} = 2.1\ \mu\text{H per phase (secondary side)},$$

referred to the primary ($\times 23.5^2 = 554$): $X_k' = 0.36\,\Omega$, $L_k' = 1.16\,\text{mH}$. During a commutation two phases are in series, so the inductance seen by the commutating loop is $2L_k = 4.2\,\mu\text{H}$; with $\sqrt{2}\,E_2 = 24\,\text{V}$ this gives the overlap angles of Table 1 and a commutation $di/dt = \sqrt{2}E_2 \sin\alpha / 2L_k \le 4\,\text{A}/\mu\text{s}$ at the thyristors (440 V, $\alpha = 36°$) – harmless. Two consequences of the very small absolute value:

- 2.1 μH is comparable with the self-inductance of a metre of secondary busbar (0.5–1 μH/m for a bar pair at typical spacing). The 6 % must therefore be specified and measured as the total from transformer terminals to bridge, and the busbar run kept short and symmetrical between the three phases, otherwise the three overlaps differ and non-characteristic harmonics appear. The transformer itself will typically contribute 4–5 %, the busbars the rest.
- If the transformer comes out with a lower reactance (3–4 % is common for a foil secondary), the difference can be made up with a small secondary reactor of 1 μH per phase, which also helps the snubber design [7].

Winding resistance. A geometric estimate for a 26 kVA core at $B = 1.5$ T with $A_{fe} \approx 60\,\mathrm{cm}^2$ gives $4.44 f B A_{fe} \approx 2.0$ V per turn, hence

$$N_2 = \frac{17/\sqrt{3}}{2.0} \approx 5 \text{ turns (star)}, \qquad N_1 = \frac{400}{2.0} \approx 200 \text{ turns (delta)}.$$

With a mean turn length $\ell_m \approx 0.5$ m, a current density of 2.5 A/mm^2 (air-cooled, class F) and copper at 75 °C ($\rho = 2.1 \times 10^{-8}\,\Omega$m):

$$A_2 = \frac{898}{2.5} = 360 \text{ mm}^2, \qquad R_2 = \rho\frac{N_2 \ell_m}{A_2} \approx 0.15 \text{ m}\Omega,$$
$$A_1 = \frac{38/\sqrt{3}}{2.5} = 8.4 \text{ mm}^2, \qquad R_1 = \rho\frac{N_1 \ell_m}{A_1} \approx 0.25\ \Omega,$$

where A_2 corresponds for instance to a 120×3 mm foil, R_2 is per secondary phase and R_1 per primary delta phase (0.083 Ω star equivalent), so that the primary resistance referred to the secondary is $0.083/554 = 0.15\,\mathrm{m}\Omega$ and the total short-circuit resistance per phase referred to the secondary is

$$R_k = R_2 + R_1' \approx 0.29 \text{ m}\Omega, \qquad e_r = \frac{\sqrt{3} R_k I_2}{E_2} = 2.7\,\%, \qquad \frac{X_k}{R_k} = 2.2.$$

The split is roughly half and half between the two windings, which is the usual optimum. Consequences:

- DC-side ohmic drop $d_r = 2R_k I_d = 0.64$ V at 1100 A (3.9 % of the output voltage) – already included in Eq. (6) and Table 1. Together with the 2.6 V of the thyristors and the 0.6 V of commutation drop, 3.9 V of the 20.5 V no-load voltage at 360 V are lost before the load, i.e. the "useful" fraction is 80 %.
- Copper losses $3I_2^2 R_k \approx 0.7$ kW at rated current (DC-resistance value); with the eddy share (10–15 % of the DC losses in a foil secondary) multiplied by $F_{HL} \approx 5$, the load losses are about 1.0–1.2 kW, plus 100–150 W of core loss: efficiency of the transformer alone about 93 %.
- The skin depth of copper at 50 Hz is 9.3 mm, so a 3 mm foil is unaffected at the fundamental; at the 11th–13th harmonic (2.3 mm) the eddy loss rises, which is what F_{HL} accounts for.
- $X/R \approx 2$ is low compared with large transformers ($X/R = 10$–20): the resistance damps the commutation and the DC-side ripple, and the secondary short-circuit current of Section 7 is limited also by e_r.

The values above are estimates for the specification: the manufacturer's short-circuit test at rated current gives e_x and e_r directly, and the DC voltage regulation should then be re-checked with Eq. (1).

V_{line} (V)	I_d (A)	V_{d0} (V)	α (deg)	u (deg)	P (kW)	Q (kvar)	PF	I_1 (A)	THD_{conv} (%)	THD_{PCC} (%)
360	1050	20.7	13.1	10.5	17.2	7.1	0.92	34.8	26.1	3.2
360	1100	20.7	12.4	11.1	18.0	7.3	0.93	36.5	25.8	3.2
400	1050	23.0	28.1	6.3	17.2	12.6	0.81	34.8	28.0	3.8
400	1100	23.0	27.8	6.7	18.0	13.1	0.81	36.5	27.8	3.8
440	1050	25.3	36.2	5.2	17.2	16.7	0.72	34.8	28.5	4.0
440	1100	25.3	36.0	5.5	18.0	17.4	0.72	36.5	28.4	4.0

Table 1: Operating points at 16.4 V output; $e_x = 6\,\%$, $e_r = 2.7\,\%$, $V_T = 1.3$ V, ratio 23.5.

4 Operating points

Table 1 gives the operating point at 360, 400 and 440 V, at 1050 and 1100 A, with the transformer of Section 3 and the drops of Section 3.2. The line current I_1 is the fundamental at the primary; THD_{conv} is the THD of the converter current (Eq. (4)); THD_{PCC} is the value after the filter of Section 5.

The reactive power is at most 17 kvar (at 440 V, $\alpha = 36°$); the displacement factor is 0.72 in that case but the absolute value is what matters on a 1 MVA connection, and the traps supply 11 kvar. No further compensation is needed.

5 AC side: line reactor and traps

5.1 Topology

The converter is a harmonic current source. The harmonic current that reaches the grid is divided between the traps and the grid path according to

$$\frac{I_{h,PCC}}{I_{h,conv}} = \left| \frac{Z_F(h)}{Z_F(h) + jh(X_S + X_s)} \right|, \tag{7}$$

where $Z_F(h)$ is the parallel of the three trap branches and X_s the line reactor. The grid reactance $X_S = 9.6\,\mathrm{m\Omega}$ is negligible against X_s at this power level, so the reactor is what makes the traps effective: without it every harmonic would split between a 0.1–0.3 Ω trap and a 0.05 Ω grid and most would go to the grid.

5.2 Line reactor

$X_s = 10\,\%$ on the converter base $S_{conv} = V_{d0} I_d = 25.3$ kVA at 400 V:

$$X_s = 0.10\,\frac{400^2}{25\,300} = 0.63\ \Omega, \qquad L_s = 2.0\ \mathrm{mH}, \qquad I_L \approx 40\ \mathrm{A\ continuous}.$$

A 2 mH / 40 A three-phase iron-core reactor. Its fundamental drop with the traps connected is small (the 16 A capacitive trap current and the converter's 15–25 A reactive current largely cancel); at no load it raises the filter bus by $Q_F X_s / V^2 = 4.4\,\%$.

5.3 Traps

Three single-tuned branches, each an L–C series with the intrinsic resistance of the reactor (quality factor $Q \approx 40$ assumed), tuned slightly below the harmonic to allow for capacitor ageing and temperature drift:

The 5th, 7th and 11th harmonics are absorbed almost entirely; what remains at the PCC is dominated by the 13th and above, which see only the line reactor (share 0.17–0.29). THD_{PCC} is

Branch	Q_C (kvar)	h_t	f_t (Hz)	C (μF)	L (mH)	R (mΩ)	I_{C1} (A)
5th	5.5	4.85	242	109	3.94	150	7.9
7th	3.0	6.80	340	60	3.67	196	4.3
11th	2.5	10.70	535	50	1.78	150	3.6

Table 2: Trap branches, star-equivalent values at 400 V ($X_C = V^2/Q_C$, $C = 1/\omega X_C$, $L = X_C/(h_t^2\omega)$, $R = h_t\omega L/Q$). I_{C1} is the fundamental current per phase.

h	5	7	11	13	17	19	23	25	35	37
$\lvert Z_F\rvert$ (Ω)	0.44	0.49	0.34	1.70	3.49	4.24	5.64	6.30	9.46	10.08
$h(X_S + X_s)$ (Ω)	3.22	4.50	7.07	8.36	10.93	12.22	14.79	16.08	22.51	23.80
$I_{h,PCC}/I_{h,conv}$	0.122	0.098	0.046	0.169	0.242	0.258	0.276	0.282	0.296	0.297

Table 3: Filter impedance, grid-path impedance and sharing factor, Eq. (7).

3.2–4.0 % over the whole line-voltage range (Table 1), and the absolute harmonic currents injected into the 1 MVA connection are below 1 A each (Table 4). If the 13th and higher must be reduced further, the 11th trap can be turned into a second-order high-pass (damping resistor across its reactor) at the cost of a few watts of fundamental loss.

Checks:

- *Parallel resonances* between the trap capacitances and the reactor: at $h \approx 3.6$ (73 Ω peak with $Q = 40$; 4.6 Ω at the 3rd harmonic), $h \approx 6.0$ (13 Ω) and $h \approx 9.1$ (20 Ω). No characteristic harmonic falls there; the non-characteristic 3rd (line unbalance) and even harmonics (firing asymmetry, see [6]) are a fraction of an ampere and produce a few volts of distortion on the filter bus at most. Keeping the firing-pulse symmetry within $\pm 0.5^\circ$ keeps the 2nd and 4th harmonics below 1 % of I_1.
- *Background distortion*: a 3 % 5th-harmonic voltage on the grid drives only 1.9 A into the 5th trap through the 3.2 Ω of the line reactor at 250 Hz – the reactor protects the traps as much as the grid.
- *Capacitors*: 480 V class minimum (525 V preferred), rated for the fundamental plus the tuned harmonic; the 5th trap carries up to 7 A of 5th harmonic on top of 8 A of fundamental.
- *Reactive balance*: converter 7–17 kvar lagging, traps 11 kvar leading; the PCC sees between -4 and $+6$ kvar. Nothing to compensate.

6 DC filter

The 6th-harmonic ripple at the bridge output, Eq. (5), is 1.4 V rms at 360 V ($\alpha = 12^\circ$) and 3.7 V rms at 440 V ($\alpha = 36^\circ$) – 23 % of the output at the worst point, so a filter is needed. With $R = 14.9\,\text{m}\Omega$ the load is so low-impedance that the reactor alone does most of the work: the transfer to the load is

$$H(j\omega_6) = \frac{1}{1 - \omega_6^2 L_{dc} C_{dc} + j\omega_6 L_{dc}/R}, \qquad \frac{\omega_6 L_{dc}}{R} = 38 \text{ for } L_{dc} = 0.3 \text{ mH}.$$

Chosen values: $L_{dc} = 0.3$ mH / 1100 A (stored energy 180 J), $C_{dc} = 5$ mF / 63 V electrolytic; resonance at 130 Hz, damping ratio $\zeta = \frac{1}{2R}\sqrt{L/C} \approx 8$ (over-damped, no ringing on load steps).

The load voltage ripple is at most 1.8 % peak-to-peak (target 5 %), and the ripple current in the reactor and bridge at most 1.8 % peak-to-peak, which validates the smooth-current assumption. The reactor guarantees continuous conduction down to any current the bridge will practically see: the critical current for discontinuous conduction is $V_6/(\omega_6 L_{dc}) \approx 6$ A. A smaller filter ($L_{dc} = 0.15$ mH, $C_{dc} = 5$ mF) would still give about 3.5 % ripple; 0.3 mH was kept for margin against the 5 % target at 440 V and for load-step damping. The capacitor ripple current is $\omega_6 C_{dc} V_{6,load} < 1$ A.

h	5	7	11	13	17	19	23	25
I_h/I_1 converter (%)	19.8	14.0	8.7	7.2	5.3	4.6	3.6	3.2
I_h/I_1 at PCC (%)	2.43	1.38	0.40	1.22	1.28	1.19	0.99	0.90
I_h at PCC (A)	0.84	0.48	0.14	0.43	0.45	0.41	0.34	0.31

Table 4: Worst-case spectrum (440 V, 1100 A, $\alpha = 36°$): converter current, current at the PCC relative to $I_1 = 36.5$ A, and in amperes. THD 28.4 % → 4.0 %.

V_{line} (V)	I_d (A)	α (deg)	V_6 bridge (V rms)	V_6 load (V rms)	$\Delta V_{pp}/V_{out}$ (%)	$\Delta I_{pp}/I_d$ reactor (%)
360	1050	13.1	1.40	0.038	0.66	0.7
360	1100	12.4	1.35	0.035	0.61	0.6
400	1050	28.1	2.75	0.075	1.30	1.3
400	1100	27.8	2.72	0.071	1.23	1.2
440	1050	36.2	3.71	0.102	1.75	1.8
440	1100	36.0	3.69	0.097	1.67	1.7

Table 5: DC-side 6th-harmonic ripple with $L_{dc} = 0.3$ mH, $C_{dc} = 5$ mF.

7 Thyristor bridge

- Per device: average current 367 A, rms 635 A; conduction loss with $V_{T0} = 0.9$ V, $r_T = 0.35$ mΩ: 470 W; 2.8 kW for the bridge, i.e. about 15 % of the output power – the price of a 16 V rectifier. Forced-air heatsinks with $R_{th,ca} \le 0.05$ K/W per device, or a common water-cooled plate.
- Voltage: repetitive peak across a blocking device $\sqrt{2} \times 17 \times 1.1 = 26$ V. A 1200–1600 V class is chosen for surge withstand and snubber energy, not for the steady state; the device is selected on current: $I_{T(AV)M} \ge$ 600–800 A capsule or two 400 A modules in parallel per position.
- Gate drive: at 360 V and $\alpha = 12°$ the anode-cathode voltage at the firing instant is only $\sqrt{2} \times 15.3 \times \sin(12°) \approx 4.5$ V, so a long or picket-fence gate pulse train is required until the current is established, with $I_G \ge 1$ A and $di_G/dt \ge 1$ A/μs.
- Protection: semiconductor fuses in the secondary legs; prospective short-circuit current at the secondary $\approx I_2/\sqrt{(e_x + X_s S_2/S_{conv})^2 + e_r^2} \approx 5.4$ kA rms, i.e. six times rated – the line reactor limits it strongly. RC snubber and MOV clamp per bridge as in [7].

8 Summary

References

[1] D. Bagnara, *Sizing of a 400 V / 1 MW thyristor AC/DC supply for a 1100 A resistive load at 16.4–200 V*, internal note, September 2026.

[2] IEC 60146-1-1, *Semiconductor converters – General requirements and line commutated converters – Part 1-1.*

[3] IEC 61378-1, *Converter transformers – Part 1: Transformers for industrial applications.*

[4] E. W. Kimbark, *Direct Current Transmission*, Vol. 1, Wiley, 1971.

[5] J. Arrillaga, N. R. Watson, *Power System Harmonics*, 2nd ed., Wiley, 2003.

[6] D. Bagnara, *Even harmonics in line-commutated converters*, internal note, 2026.

Item	Rating
Line	400 V ±10 %, $S_n = 1$ MVA, $S_{sc} \approx 16.7$ MVA (assumed)
Line reactor L_s	2.0 mH (0.63 Ω, 10 % on 25.3 kVA), 40 A, three-phase iron core
Traps	5th: 5.5 kvar, 109 μF, 3.94 mH; 7th: 3.0 kvar, 60 μF, 3.67 mH; 11th: 2.5 kvar, 50 μF, 1.78 mH; capacitors 525 V class; total 11 kvar
Transformer	26.4 kVA, 400 V Δ (optional ±2 × 2.5 %) / **17 V**, 898 A, $e_x = 6$ % ($L_k = 2.1$ μH/phase secondary, 1.2 mH primary), $e_r = 2.7$ % ($R_k = 0.29$ mΩ/phase referred to secondary), $P_{cu} \approx 0.7$ kW + eddy, converter duty IEC 61378-1, $F_{HL} \approx 5$
Bridge	six-pulse fully controlled, 1100 A, 1600 V class thyristors $I_{T(AV)} \geq 600$ A, $\alpha = 12$–36° over the line range
DC filter	$L_{dc} = 0.3$ mH / 1100 A, $C_{dc} = 5$ mF / 63 V
Performance	16.4 V / 1100 A from 360 to 440 V; $Q \leq 17$ kvar; Q_{PCC} within ±6 kvar; $\mathrm{THD}_{PCC} \leq 4.0$ % (own fundamental, 36.5 A); load ripple ≤ 1.8 % pp

Table 6: Design summary.

[7] D. Bagnara, *Snubber design for diode and thyristor bridges*, internal note, 2026.